

Qatar-1b

R-0215

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_181121 · created 2026-07-17T00:37:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458443.6608 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.084 +/- 0.042
Transit depth (Rp/Rs)^2	0.7 +/- 0.7 [%]
Orbital Inclination (inc)	88.95 +/- 3.13
Airmass coefficient 1 (a1)	1.316 +/- 0.024
Airmass coefficient 2 (a2)	-0.156 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.84 %
Best Comparison Star	None
Optimal Aperture	2.71
Optimal Annulus	8.8
Transit Duration (day)	0.0741 +/- 0.0083

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.084 ± 0.042 vs 0.1463 (deviation 1.09σ)
Duration fit vs published	0.0741 d vs 0.0692 d (deviation 0.44σ)
Mid-time O–C	-0.5 min
Depth SNR	1.5
Residual scatter	1.84 %

Light curve

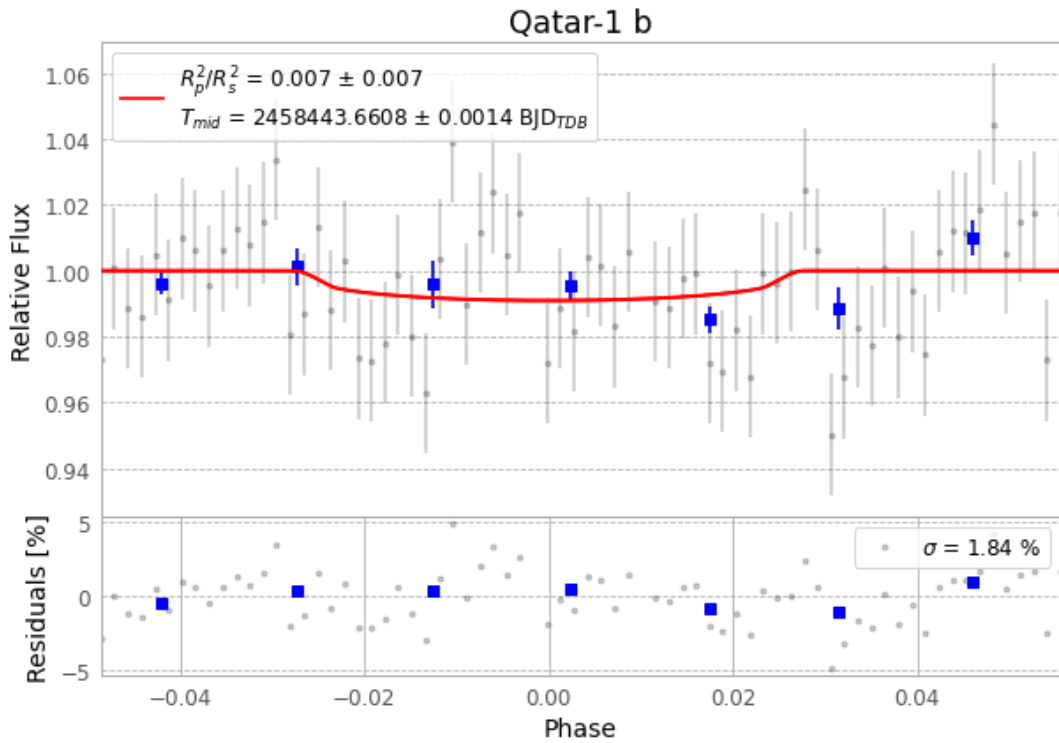


Figure 1 — FinalLightCurve_Qatar-1 b_2018-11-20.png (SHA-256 98d8f5174cdbd9d0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [376, 411], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6d00d6ceebe644dd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

72 FITS frames (47 MB), Qatar-1181121020912.FITS → Qatar-1181121054212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-181121.log (b138b6a9a109...), FinalLightCurve_Qatar-1 b_2018-11-20.png (98d8f5174cdb...), FinalLightCurve_Qatar-1 b_2018-11-20.csv (a34003239679...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 00:37Z.